SPECIAL ARTICLE

Wearable Technology and Its Role in Neurologic Care

Emerging Issues in Neurology

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Abstract

Consumer wearable devices are commonly used by patients and consumers for several reasons with increasing application as new technologies are developed. Use of these devices is an emerging issue in Neurology because of increased adoption and the additional data reported to providers by patients. Understanding of possible functions, limitations, and effect on patients of non–US Food and Drug Administration (FDA)-cleared wearable technology to inform neurologic care is needed. A common theme in people with neurologic conditions regarding consumer wearables and associated tracking applications is that there is significant promise in these tools, but adherence (days per use/per week), continued engagement (attrition), and unintended consequences such as heightened anxiety remain important issues. Further understanding and validation of these devices is needed within the field of Neurology before full use and confidence can be achieved. Below, we provide examples of non–FDA-cleared wearable devices used in Neurology in the areas of epilepsy, headache, cardiac monitoring, and sleep.

Introduction

Wearable devices, defined as electronics worn to monitor activities and vital signs,[1] are increasingly popular for health care purposes and are emerging in neurologic care.[2] These consumer devices can offer valuable data for health care providers and are becoming part of daily activities and life for many[3]; however, their use and applicability in neurology are not widely studied.[4] Misunderstanding proper use and interpretation by consumers, patients, or medical providers is a barrier to formal health care use and may lead to elevated anxiety in patients, inappropriate diagnosis, and/or cost.[3] Scalability, cost-related disparities in care, patient protection, false positives, and integration with electronic health records are among existing concerns.[1,4-7] Despite these challenges, the use of consumer wearables is expected to increase, potentially improving the quality of patient data and access to acute medical changes. This study aims to provide a high-level view of the use, limitations, and evidence for non–US Food and Drug Administration (FDA)-cleared digital and wearable technology in neurologic care, providing examples of its use in some fields of Neurology. FDA authorization of devices does not exist in a manner consistent with pharmaceutical drug approval. Rather, a device can be FDA cleared, approved, or authorized. FDA authorization is a process that involves review for significant safety and effectiveness of a medical device.[8] Devices that carry FDA authorization are not within the scope of this article, and the focus was to provide discussion/commentary on how providers can think about use of commercial-grade devices purchased by a patient either on advice of a medical professional or on their own initiative.

Scope and Disclaimers

Emerging Issues in Neurology (EIN) articles are published by the American Academy of Neurology (AAN) and its affiliates. The goal of the EIN series is to provide timely and informal

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Glossary

AAN = American Academy of Neurology; **AFib** = atrial fibrillation; **EIN** = Emerging Issues in Neurology; **FDA** = US Food and Drug Administration; **PPG** = photoplethysmography.

guidance (derived from expert consensus opinion) to neurologists about new or emerging issues that have immediate implications for patient care, but for which the formal evidence base is still evolving. The information in an EIN article is provided as an educational service and (1) should not be considered inclusive of all proper treatments or methods of care, (2) does not represent a statement of the standard of care, (3) is not continually updated and may not reflect the most recent evidence, (4) does not mandate any specific course of medical care, (5) represents the medical/scientific opinion of the author panel, (6) is not the result of a systematic review, and (7) is not intended to replace the independent professional judgment of the treating provider. In all cases, decisions about patient care should be considered in the context of treating the individual patient. Use of the information is voluntary. The AAN provides this information on an "as is" basis, and makes no warranty, expressed or implied, regarding the information. The AAN specifically disclaims any warranties of merchantability or fitness for a particular use or purpose. The AAN assumes no responsibility for any injury or damage to persons or property arising out of or related to any use of this information or for any errors or omissions.

Case Examples

Heart Rate and Cardiac Arrhythmia Screening

Smart watches use the technology of photoplethysmography (PPG) to measure heart rate. This technique measures volumetric changes of the heart by measuring light transmission or reflection. PPG can be seen in physical sensors that are medical grade as well as consumer devices such as smart watches or phone apps. Software algorithms are used to extrapolate heart rate from the PPG data. Some commercially available devices, such as Kardia and Apple Watch, use proprietary algorithms and medical-grade sensors that are FDA-cleared to detect irregular heart rhythms. However, not all devices and not all device features meet that standard. PPG can underestimate heart rate in sinus tachycardia, and its accuracy may suffer during physical activity.[9] Approximately 25% of ischemic strokes are due to atrial fibrillation (AFib) with stroke often being the presenting symptom.[10] AFib increases an individual's risk of having a stroke 5-fold.[11] Many smart watches have programs to alert consumers to the possibility of the presence of AFib or an irregular pulse. The Apple Heart Study found that in the 419,297 participants 0.52% of participants received irregular pulse notifications. A total of 450 people (20.8% of those people notified) went on to receive an ECG patch testing and 153 people (34%) had recorded AFib. In addition, the study went on to report that 44% of those people who were notified about an irregular rhythm went on to report a new diagnosis of AFib compared with only 1.0% of participants who received no irregular rhythm alerts being diagnosed with AFib. Similarly high positive prediction values for AFib were found in the Fitbit Heart Study.[12] Limitations to both studies include a younger average age of participants with lower levels of comorbidities such as hypertension and diabetes than the general population. The younger age of device users who in turn have a lower risk of AFib was a concerning limiting factor because it may mask the tendency of devices to create false positive and unnecessary testing. In the Apple Heart Study, adverse events were reported in 16 people, with the most common adverse event being anxiety in 15 people.[13] The Fitbit Heart Study reported adverse effects of skin irritation in 0.28% of patients and stress or anxiety in 0.25% of patients. One in 5 patients with AFib reported anxiety related to alerts in a recently published *Journal of the American Heart Association* study.[14]

Handheld ECG devices attached to a smartphone, such as the brand name Kardia, allow patients to take ECG readings outside of the physician's office or hospital. These devices can vary from 1 lead up to 12 lead ECG reads. Some brands are FDA-cleared for detecting AFib and certain heart conditions (Kardia, Heartcheck) whereas others are not (Checkme, Yongrow, Emay). The ECG recording could be 1 lead or 6 leads, not the standard 12 leads obtained in the medical setting. A single-lead ECG can miss ST segment elevation, ischemic changes in anterior, lateral and inferior walls, or lateral ventricle hypertrophy.[15] Six-lead ECG devices have been shown to be superior to a single-lead device in diagnosing AFib.[16] An in-office 12-lead EKG would still be recommended if a patient presents with an abnormal reading from a consumer device to confirm findings.

Overall, smartwatches and smart ECG devices can serve as a screening tool. A physician may wish to pursue confirmatory medical grade testing if condition is unexpected but should consider treating to prevent harm while awaiting diagnostic confirmatory testing. Clinicians should also recognize that a patient can have significant anxiety related to wearable device alerts or recordings. The device utilization may help some patients but cause harm to others and recommendations to avoid use in hypervigilant patients with otherwise normal testing may be considered. Because of lack of evidence-based literature, no guidelines are available to offer further treatment guidance to clinicians.

As technologies advance, future opportunities will emerge through the development of more sophisticated rechargeable batteries and nanotechnologies. These innovations will enable smaller, longer-lasting devices that can be worn discreetly and comfortably, even becoming implantable, without compromising compatibility with other medical tests such as MRIs. Incorporation of additional data sensors such as troponin detectors with nanotechnology may allow for detection of possible myocardial damage alerts. AI-informed algorithms will only develop more precision in diagnosis and avoidance of inaccurate alerts.[17]

Epilepsy

Consumer wearable devices and associated digital applications play various roles in epilepsy management, including seizure detection, understanding seizure effect, identifying triggers, forecasting seizure risk, symptom logging, and managing comorbidities. The most studied application is detecting convulsive seizures to alert caregivers, reducing morbidity and mortality.[18] Currently, there is one FDA-cleared wearable for detecting tonic-clonic seizures during rest and several European Conformity marked devices in Europe. These wrist or arm-worn devices use sensors for accelerometry, PPG, and electrodermal activity with proprietary algorithms in the associated application to detect motor, heart rate, and autonomic changes associated with tonic-clonic and other major motor seizures, and have been well-validated against gold standard video-EEG monitoring.[19] Similar sensors in consumer devices, like smartwatches and fitness trackers, are also used for seizure detection.[20] Few of the algorithms used by these consumer devices have undergone validation studies, comparing performance to the video-EEG gold standard, and reporting sensitivity and false alarm rates, but some have performance comparable with purpose-built devices for detecting convulsive seizures[21,22] and 1 smartwatch application recently received FDA clearance for tonic-clonic seizure detection in patients age ≥5 years.[23] Patients have expressed a preference for using consumer wearables for epilepsy monitoring as they are less stigmatizing and attract less attention than specialized wearables.[24]

Standard features of consumer devices such as smartwatches and other heart rate monitoring devices can also provide clues to seizures. This is important for neurology providers because many people with epilepsy are unaware of their seizures, making it difficult for neurology providers to make clinical decisions such as medication changes or providing driving clearance based solely on self-report.[25] For instance, most clinically significant seizures, including focal impaired aware seizures and generalized tonic-clonic seizures, are associated with a rapid increase in heart rate of >10 bpm or ≥20% from baseline.[26,27] Unexplained changes in logged heart rate, especially when coupled to other features, could be a clue to the patient and their clinician that there was an unrecognized seizure. For example, a case report documented a likely focal impaired aware seizure in a patient on a cycling tour, identified by Global Positioning System tracker data and heart rate elevation.[28] Preliminary studies have used physiologic signals from consumer wearables, including heart rate, sleep, and activity, to develop machine learning models to forecast seizure risk in the next 24 hours but clinic-ready applications are still unavailable.[29-31]

Consumer wearable devices and associated applications also aid in self-management of epilepsy and comorbidities. Evidence-based programs for disease education, symptom management, or decision support tools can be delivered through tablets and smartphones.[32] Mobile electronic diaries for tracking seizures and symptoms provide valuable feedback to patients and neurologists. A recent study using a smartwatch diary was used to identify self-reported seizure triggers in 234 patients, identified stress and poor sleep as common seizure triggers, which could be targeted with tailored interventions.[33] Data from electronic seizure diaries can identify patterns in seizure occurrence, useful for forecasting high-risk periods.[34] Some preliminary studies have demonstrated that integration of a consumer fitness tracker with an electronic seizure diary improves the accuracy of seizure forecasting.[30] If these findings are confirmed, wearable technology may lead to personalized and dynamic therapeutic options for people with seizures.

Headache

Wearables have been studied in headache management for provision of Biofeedback with some receiving clearance by the FDA. Noncleared FDA mimic devices are also available to purchase. Mimic devices do not undergo the same FDA clearance. Biofeedback, a Grade A evidence-based treatment,[35] has been studied using wearable sensors for people with migraine in both the pediatric and adult populations, that is, Cerebri and HeartMath.[36-38] Preliminary research suggests that personal consumer devices attached to wearable sensors allow for using self-delivery of biofeedback either through electromyographic voltage, finger temperature, or heart rate variability measurement. Some studies (citation 48) showed that engagement was an issue, but those who engaged with the heart rate variability sensor had clinically significant improvements in migraine related quality of life.

In addition, actigraphy has been used to better understand the everyday movements of those with headache disorders. In a study of people with various headache types who were asked to complete emailed daily headache diaries and to use a daily Fitbit device,[39] when a headache was present, participants slept more, had less physical activity, and had a lower maximum heart rate. In this study, participants were required to already own a Fitbit. These were also people who, on average, were more active than the regular population, thus limiting generalizability. Complete data collection is challenging with 509 participants enrolled in the study, yet 68.7% (350/509) of participants had what the researchers termed "dense survey" and activity data, and just 55.6% (283/509) of participants had dense survey and heart rate data. In a much smaller study of 4 patients with chronic cluster headache,[40] the Empatica E4 device was used for actigraphy recordings on the nondominant wrist to determine the amount of physiologic movement. Of the 15 of 34 attacks which met eligibility criteria, a decrease in energy expenditure was found in the preictal headache period and during the attack. This finding was counter to the typical stereotype of patient pacing because of pain.

Some of these wearables also include headache apps, for headache tracking. Future work should focus on implementing the use of wearables into clinical practice and how remote monitoring can be introduced to allow for the clinicians to monitor use and potential benefit.[41]

Sleep

Consumer wearable technologies for measuring sleep duration and depth have proliferated significantly over the past several years. Consumer wearables include single-sensor wristwatches or bands and finger-worn ring devices, predominantly based on accelerometry. These devices are similar to medical-grade activity monitors that are extensively used in sleep medicine for sleep quantification and evaluation of circadian disorders. More recently, multisensor devices typically incorporate accelerometry together with PPG to index pulse rate variability (which is tightly coupled with wake-sleep state and sleep stage depth), and variable application of other sensors for monitoring oximetry, skin conductance, and ambient light. There are also a few electroencephalography-based headband devices that provide even greater fidelity for sleep stage and architecture estimation.

In general, most consumer sleep detection devices have been reasonably well-validated against the gold standard of laboratory polysomnography (i.e., in-laboratory sleep studies which incorporate necessary EEG, electrooculography, and chin electromyography sensors to fulfill American Academy of Sleep Medicine standards for accurate sleep stage scoring). Most consumer sleep wearables have demonstrated good fidelity in healthy adult populations for detecting sleep-wake duration (i.e., total sleep time, sleep efficiency calculated as the total sleep time divided by time in bed) and sleep stage estimation for light non-rapid eye movement sleep,[42,43] as well as deep sleep.[44,45] However, common limitations for most current consumer devices are an underestimation of wake after sleep onset time duration even in healthy adults, and lack of validation in neurologically and medically diseased populations, and those with clinical sleep disorders (such as sleep disordered breathing, insomnia, hypersomnia, parasomnia, and sleep-related movement disorders). A further limitation is that most consumer sleep wearables do not provide raw data accessibility to enable review for confirmatory sleep-wake state analyses, and these devices use proprietary algorithms for sleep-wake estimation, giving sleep clinicians and researchers pause toward adopting consumer technologies without ability to scrutinize validity of accurate sleep-wake state detection.

However, despite these limitations, consumer sleep wearables will likely continue to expand in their utility and acceptance. This is because of the advantages these devices afford for scalability to large population evaluations, as well as their lower cost and convenience. These devices offer comfortable, naturalistic evaluation of sleep for people in their customary home sleep environments without the need for expensive, resource-intensive, and scarce in-laboratory sleep evaluations.

There have been some recent proof-of-concept studies suggesting that consumer sleep wearables may have utility for improving public sleep health priorities. These studies indicate that wearables can help improve sleep quality and modify undesirable sleep behaviors in healthy populations.

Future opportunities include extending such applications to diseased populations and incorporating consumer wearables into the clinical evaluation and longitudinal follow-up of a diversity of sleep disorders. This is especially relevant for insomnia and circadian sleep-wake patient populations, for monitoring outcomes of standard recommended behavioral sleep health interventions. In addition, it can assure sleep improvements in those with sleep disordered breathing receiving positive airway pressure and other interventions. A frontier in this field is the diagnosis and monitoring of treatment outcomes for sleep-related movement disorders and parasomnias.

Discussion

The landscape for consumer devices offering health screening, monitoring, and even treatments is rapidly changing and multiplying. Consumer devices are directly marketed to the public without any official entity review of safety and effectiveness; therefore, reimbursement is unlikely and performance may have more variability in interpretation for utility. Neurologists will be working with some patients who will prefer to use consumer devices because of immediate access, affordability, less involvement of insurance companies, ease of use, lack of stigma, and direct access to their personal data. There is great potential for this technology to provide a more complete and consistent data set for each patient. This would allow for greater insights into their personal level of disability from neurologic disease and possible identification of triggers to allow for a more individualized and successful treatment plan.

In this time of rapid advancement, acknowledgment and discussion with patients regarding limitations is important. Conversations regarding possible false positives and negatives will become an important part of patient education. When do we need to replicate tests with medical grade equipment to confirm suspected diagnosis rather than decide to treat with consumer technology reports will be an important decision made by neurology providers. If tests are replicated, it may mean higher health care expenses and a delay in receiving medical treatment which poses 1 set of risks to a patient. Alternatively, only treating after consumer device testing has been completed may lead to unnecessary interventions that could cause harm to patients in the form of side effects or expense of treatments. If a patient's compliance with device use is not evaluated, it may lead physicians to misjudge the level of disability or disease-related risk because of incomplete data.

Common themes in people with neurologic conditions regarding digital tracking and wearables is that there is significant promise in these tools, but adherence (days per use/per week) and continued engagement (attrition) remain important issues. A recent study showed that certain adherence enhancements such as the principles of behavioral economics, digital navigators, and neuroscience education therapy can improve adherence to diary and smartphone based behavioral therapy.[46] In addition, the increased data may lead some patients to higher levels of anxiety and worry surrounding their risk of health problems. Physicians will need to counsel around use of these devices knowing the individual using the technology.

Data privacy regarding what data are being collected by the device, where the data are stored and for how long, and who has access to the information, are all important issues that need to remain top of mind if a neurologist is recommending a consumer product. The health data privacy laws in the United States are different from other countries and patients should be advised of their personal risk if this information is not readily available by the proprietor of the technology.

As these technologies continue to proliferate, neurologists will need to become familiar with their capabilities and limitations. Medical professionals will be looked to by their patients to know when it will be worth spending money, to assess value, and risk.

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S.M. Benish: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. D. Friedman: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S. Merchant: drafting/revision of the manuscript for content, including medical writing for content. M.T. Minen: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. E.K. St. Louis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A.D. Patel: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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